LOCAL COMPLETE INTERSECTION FOR FINITE TYPE MAPS VIA COMPLETED LOCAL FACTORS

THE STACKS PROJECT AUTHORS

Editorial note. The proof ends 'Some details omitted': the omitted content is the localization bookkeeping reducing (1) to a Koszul-regular presentation (Koszul regularity is local on the neighbourhood, per the cited More on Algebra section). Declared dependency: lemma-well-defined-if-you-can-find-good-factorization is not carded; recorded via external_references.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 2523 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Good factorization criterion. lemma-well-defined-if-you-can-find-good-factorization (same chapter, named; NOT carded – build-gate exclusion). Transfers between the kernel condition and completed CI homomorphisms; declared as an external reference, not re-proved (its source proof is omitted).
- Regular sequences in neighbourhoods. algebra-lemma-regular-sequence-in-neighbourhood (cross-chapter, named). Localizes the regular-sequence condition on the presentation ideal.

Source reference. Original reference 061L: focused-seeds/geometry/S9-dpa-finite-type-lci-map/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 09QD: sources/stacks-dpa/pinned-original-source.tex, source lines 2467–2467. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07D0: focused-seeds/geometry/S9-dpa-finite-type-lci-map/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 09CC: focused-seeds/geometry/S9-dpa-finite-type-lci-map/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

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proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let A be a Noetherian ring. Let $A \to B$ be a finite type ring map. The following are equivalent*

(1) *$A \to B$ is a local complete intersection in the sense of More on Algebra, Definition 07D0,*
(2) *for every prime $\mathfrak{q} \subset B$ and with $\mathfrak{p} = A \cap \mathfrak{q}$ the ring map $(A_\mathfrak{p})^\wedge \to (B_\mathfrak{q})^\wedge$ is a complete intersection homomorphism in the sense defined above.*

Proof. Choose a surjection $R = A[x_1, \ldots, x_n] \to B$. Observe that $A \to R$ is flat with regular fibres. Let I be the kernel of $R \to B$. Assume (2). Then we see that I is locally generated by a regular sequence by Lemma 09QD and Algebra, Lemma 061L. In other words, (1) holds. Conversely, assume (1). Then after localizing on R and B we can assume that I is generated by a Koszul regular sequence. By More on Algebra, Lemma 09CC we find that I is locally generated by a regular sequence. Hence (2) hold by Lemma 09QD. Some details omitted. □